

Chapter 11

Review for Midterm 1

Lecture 11 | Devore 9e: Ch. 2–3.5

Midterm 1 covers Lectures 1–10: probability spaces, counting, conditional probability, independence, Bayes, and discrete random variables through variance.

Remark 11.1 (Where the line actually fell, and it moved). The two semesters did not cut at the same place, so decide this before writing the exam. These notes follow the Fall 2025 sequence, in which Lecture 10 (variance, binomial, geometric, hypergeometric) came *before* the exam. The archived Spring 2026 paper (`exams/2026-spring/tex/mid1.tex`) stopped one topic earlier — its six problems end at “Random Variable, PMF, CDF, and Expected Value”, with no variance and no named families, because that semester ran two lectures behind at this point. Sections §2 and §3 below cover the wider Fall scope; trim them if your pace matches Spring’s.

11.1 What you must be able to do

1. **Write down a sample space** and express a verbal event as a subset (Lecture 2). Translate “at least one”, “none”, “exactly two” into unions, complements, intersections.
2. **Count** (Lecture 3): decide *order?* and *replacement?*, then use the right entry of the table.
3. **Compute geometric probabilities** as ratios of length/area (Lecture 4).
4. **Use the probability axioms** (Lecture 4) to prove small identities: $\mathbb{P}(A^c) = 1 - \mathbb{P}(A)$, $\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B)$, inclusion–exclusion.
5. **Condition** (Lectures 5–7): multiplication rule for sequential experiments, law of total probability when there is a hidden case split, Bayes when the question reverses the given.
6. **Check independence** from the definition $\mathbb{P}(A \cap B) = \mathbb{P}(A)\mathbb{P}(B)$ — and know that pairwise does not imply mutual (Lectures 5–6).
7. **Work with a discrete random variable** (Lectures 8–10): find the pmf, build the cdf, compute $\mathbb{E}[X]$, $\mathbb{E}[h(X)]$, and $\text{Var}(X)$.
8. **Recognize the named families** so far: Bernoulli (the indicator variable of Lecture 8), binomial, geometric, hypergeometric.
9. **Find an expected count with indicators** (Lectures 9–10): write the count as a sum of indicators and add the probabilities, whether or not the events are independent; and set up a first-step equation when a game can restart (Lecture 6).

11.2 Formula sheet

Counting

	without replacement	with replacement
order matters	$A_n^k = \frac{n!}{(n-k)!}$	n^k
order doesn't matter	$\binom{n}{k}$	$\binom{n+k-1}{n-1}$

Probability

$$\mathbb{P}(A^c) = 1 - \mathbb{P}(A) \qquad \mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B)$$

$$\mathbb{P}(A \mid B) = \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)} \qquad \mathbb{P}(A \cap B) = \mathbb{P}(A)\mathbb{P}(B \mid A)$$

$$\mathbb{P}(B) = \sum_i \mathbb{P}(A_i)\mathbb{P}(B \mid A_i) \quad \mathbb{P}(A_i \mid B) = \frac{\mathbb{P}(A_i)\mathbb{P}(B \mid A_i)}{\sum_j \mathbb{P}(A_j)\mathbb{P}(B \mid A_j)}$$

$$A \perp\!\!\!\perp B \iff \mathbb{P}(A \cap B) = \mathbb{P}(A)\mathbb{P}(B). \text{ Odds form: } \frac{\mathbb{P}(A \mid B)}{\mathbb{P}(A^c \mid B)} = \frac{\mathbb{P}(A)}{\mathbb{P}(A^c)} \cdot \frac{\mathbb{P}(B \mid A)}{\mathbb{P}(B \mid A^c)}.$$

Discrete random variables

$$F_X(x) = \mathbb{P}\{X \leq x\}, \quad \mathbb{E}[X] = \sum_i x_i p_i, \quad \mathbb{E}[h(X)] = \sum_i h(x_i) p_i,$$

$$\text{Var}(X) = \mathbb{E}[(X - \mu)^2] = \mathbb{E}[X^2] - \mathbb{E}[X]^2, \quad \sigma_X = \sqrt{\text{Var}(X)},$$

$$\mathbb{E}[aX + b] = a\mathbb{E}[X] + b, \quad \text{Var}(aX + b) = a^2 \text{Var}(X), \quad \sigma_{aX+b} = |a| \sigma_X.$$

$$\mathbb{E}[X_1 + \dots + X_n] = \mathbb{E}[X_1] + \dots + \mathbb{E}[X_n] \text{ (no independence needed),} \quad \mathbb{E}[\mathbf{1}_A] = \mathbb{P}(A),$$

$$\mathbb{E}[N] = \sum_{k \geq 1} \mathbb{P}\{N \geq k\} \text{ for } N \text{ with values in } \{0, 1, 2, \dots\}.$$

Families so far

Name	pmf	support	$\mathbb{E}$	Var
Bernoulli(p)	$p^k(1-p)^{1-k}$	$k \in \{0, 1\}$	p	$p(1-p)$
Binomial(n, p)	$\binom{n}{k} p^k (1-p)^{n-k}$	$k = 0, \dots, n$	np	$np(1-p)$
Geometric(p)	$(1-p)^{k-1} p$	$k = 1, 2, \dots$	$\frac{1}{p}$	$\frac{1-p}{p^2}$
Hypergeometric(n, M, N)	$\frac{\binom{M}{k} \binom{N-M}{n-k}}{\binom{N}{n}}$	k feasible	$n \frac{M}{N}$	$\frac{N-n}{N-1} n \frac{M}{N} \left(1 - \frac{M}{N}\right)$

11.3 Five quick checks

Each should take about a minute. Answers follow.

1. What is the probability of at least one six in four rolls of a die?
2. Two dice are rolled. Given that at least one of them shows a 1, what is the probability that the sum is 7?

3. On average, how many different faces appear in six rolls of a die?
4. A coin is tossed 10 times. On average, how many times do two consecutive tosses both come up heads? (In HHH, count two.)
5. What is the variance of a single roll of a fair die?

Answers.

1. By the complement, $1 - (5/6)^4 = \frac{671}{1296} \approx 0.518$ — slightly better than even.
2. The condition leaves the 11 outcomes containing a 1; of these (1, 6) and (6, 1) sum to 7. Answer $\frac{2}{11}$.
3. One indicator per face: face j appears with probability $1 - (5/6)^6$, so the expected number is $6(1 - (5/6)^6) \approx 3.99$. Six rolls typically miss two faces.
4. One indicator for each of the 9 adjacent pairs, each HH with probability $\frac{1}{4}$: the answer is $\frac{9}{4}$. Linearity ignores the overlaps.
5. $\mathbb{E}[X] = \frac{7}{2}$ and $\mathbb{E}[X^2] = \frac{91}{6}$, so $\text{Var}(X) = \frac{91}{6} - \frac{49}{4} = \frac{35}{12} \approx 2.92$.

11.4 The three mistakes that cost the most points

1. **Confusing $\mathbb{P}(A \mid B)$ with $\mathbb{P}(B \mid A)$.** Read the sentence again: what is *given*, and what is being *asked*? The disease example (Lecture 7) exists entirely to make this vivid.
2. **Treating disjoint as independent.** Disjoint events with positive probability are maximally *dependent*. “Mutually exclusive” is a statement about $A \cap B$; “independent” is a statement about $\mathbb{P}(A \cap B)$.
3. **Forgetting that $\mathbb{E}[h(X)] \neq h(\mathbb{E}[X])$.** In particular $\mathbb{E}[X^2] \neq \mathbb{E}[X]^2$; their difference is the variance.

Remark 11.2. Practice problems for this midterm are in `exams/2026-spring/tex/` — see `practice1.tex` and `make-up-exam.tex`, both with the same topic coverage. (`mid2.tex` and `mid2-makeup.tex` are Midterm 2 material.)